Statistical Learning with Regression Models Project

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1 Introduction

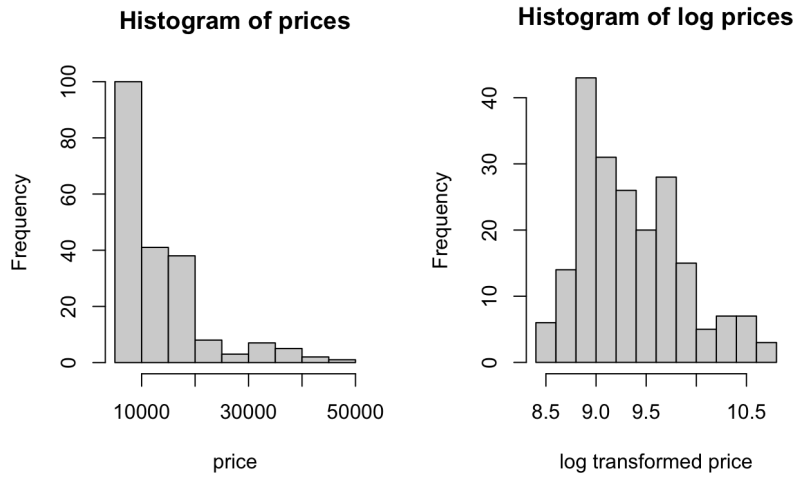
In this report, which is consisting of two parts, we are first looking at a data set containing information on a number of cars in the market during the 80′s. Some of the information that we have access to is for example car dimensions (width, length, height), number of cylinders of the car, the horsepower and other specs, as well as the price of the car. Our problem is to, via various methods and analyses, find a multivariate linear model which will to a satisfactory degree describe the price of a car based on some selected covariates.

In the second part we study a data set concerning air traffic controllers (ATC) "number of errors" which have been committed. The goal is to construct a GLM which can predict the rate of errors based on some covariates.

2 Linear regression for cars' data

2.1 Methods

We begin by looking at a histogram of the response variable *price*. In figure 1a we see a right skewed distribution of the prices in the data, hence signaling that a log transformation might be useful. The effect of such a transformation is shown in figure 1b where we now see a more symmetrical distribution. Hence it is of interest to use a log transformed response variable for our final model and we assume moving forward that our analysis is performed using this transformation. Note that when we say log transformation we are using the natural logarithm ln, but this could be chosen as any logarithmic function with arbitrary base.



(a) Histogram of the prices in the dataset

(b) Histogram of the log transformed prices in the dataset

Figure 1: Two histograms showing the effect of log transforming the price distribution.

Now we look at the continuous covariates. We immediately draw some conclusions about irrelevance and redundancy of some of the covariates just by their description. For example will *car_ID* trivially not be relevant to study as it is arbitrarily chosen numbers labeling the data. We also note that redundancy between *citympg* and *highwaympg* is to be expected as fuel usage will likely not vary much whether the car is driven in the city or on the highway (see Appendix 4).

In figure 2 we note of some continuous covariates which might be of interest with respect to the response variable *price*, these being *wheelbase, carlength, carwidth, curbweight, enginesize, horsepower, citympg* and *highwaympg*. Disregarding the redundant covariate *highwaymph* (see Appendix 4) we look for potential multicollinearity between the remaining ones. For example we may look at a larger model containing the previously noted covariates and a smaller model composed of only *curbweight, horsepower* and *citympg*, meaning

$$E[price|\dots] = \beta_0 + \beta_1 \cdot curbweight + \beta_2 \cdot horsepower + \beta_3 \cdot citympg, \qquad (1)$$

where β_0 is the intercept of the model and $\beta_i, i = 1, 2, 3$ are the corresponding coefficients of each covariate.

The selection of covariates for the smaller model can be motivated looking at which numerical covariates is selected by Olsrr for each number of allowed covariates and by analyzing the variance inflation factors between the covariates (see appendix 4). Performing a partial F-test between the models, we observe

in figure 3 that the larger model does not significantly improve the model fit over the smaller one. For example, we get a p-value of $p = 0.058 > 0.05$ and we are therefore unable to reject the smaller model (1).

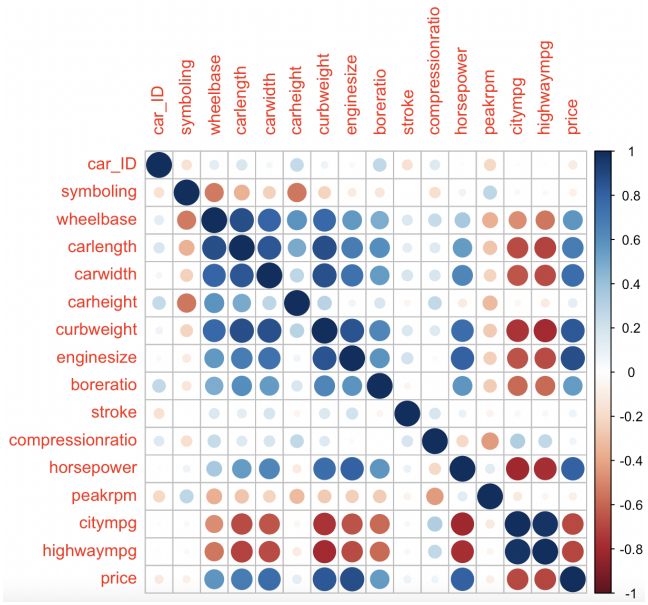


Figure 2: Pairwise correlation plot of continuous covariates.

```
Model 1: log(car_price_data$price) ~ car_price_data$curbweight + car_price_data$horsepower +
    car_price_data$citympg
Model 2: log(car_price_data$price) ~ car_price_data$curbweight + car_price_data$horsepower +
    car_price_data$citympg + car_price_data$wheelbase + car_price_data$carlength +
    car_price_data$carwidth + car_price_data$enginesize
  Res.Df    RSS Df Sum of Sq     F  Pr(>F)
1    201 7.7068
2    197 7.3592  4   0.34756 2.326 0.05777 .
```

Figure 3: A partial F-test performed between a larger model constructed of seemingly interesting covariates from a correlation plot and a smaller model.

Now, looking at the categorical covariates, we reason that the categorical covariate *carbody* (see appendix 4) seem relevant in describing the price, and as such we try adding it to our model (1). Using *carbodyhatchback* as the baseline we get the multivariate linear model

$$E[\log(price)|\dots] = \beta_0 + \beta_1 \cdot curbweight + \beta_2 \cdot horsepower + \beta_3 \cdot citympg + \sum_i \beta_{4,i} \cdot carbody_i, \tag{2}$$

where

$$\beta_{4,i} = \begin{cases} \beta_{convertible} & \text{if } carbody = convertible, \\ \beta_{hardtop} & \text{if } carbody = hardtop, \\ \beta_{sedan} & \text{if } carbody = sedan, \\ \beta_{wagon} & \text{if } carbody = wagon, \end{cases}$$

and

$$carbody_i = \begin{cases} 1 & \text{if } carbody = i, \\ 0 & \text{otherwise.} \end{cases}$$

We are now interested to check if our model perhaps is unnecessarily complex and if perhaps some covariate could be removed from the model. To do this we perform backward search on the model (2). In table 1 we see that AIC (Akaike's information criterion) decreases when we remove *citympg* and increases for all other models where we remove a covariate, we also have a very small p-value in each of these cases. Hence this is some motivation that we should with significant probability disregard *citympg* as it makes the model unnecessarily complex. We therefore propose the following model

$$E[\log(price)|\dots] = \beta_0 + \beta_1 \cdot curbweight + \beta_2 \cdot horsepower + \sum_i \beta_{3,i} \cdot carbody_i. \tag{3}$$

Removed covariate	Df	Sum of Squares	RSS	AIC	F value	$\Pr(>F)$
citympg	1	0.0500	6.3897	-697.01	1.5525	0.2142
none (full model)	–	–	6.3397	-696.62	–	–
carbody	4	1.3670	7.7068	-664.59	10.6197	8.196×10^{-8}
horsepower	1	1.2760	7.6157	-661.02	39.6489	1.939×10^{-9}
curbweight	1	5.6452	11.9850	-568.07	175.4190	$< 2.2 \times 10^{-16}$

Table 1: Backward search for log(price)

Now that we have a model we want to look at potential outliers with respect to our model. An in depth analysis can be found in the appendix 4. We there note that some of the influential observations which could be found correspond to premium or abnormally priced vehicles, reflecting brand or technological effects which our model does not consider. We thus decide to keep them as by removing them we are making our data and conclusions biased towards more "average" cars.

Something which is yet to be explored is the predictive ability of our model. This is of interest as the model might perform well on the details of our observed data but be overfitting, meaning it may not generalize well to new observations. Hence, the model might have high variance, and the results could vary a lot between similar datasets.

One way we could assess model bias by splitting the data into a training set with 70% of the original data and a test set with 30%. We then use *regsubsets* to

identify, for each model size, the subset of covariates that minimizes the residual sum of squares on the training data. Each model is then evaluated by testing its performance on the test data and calculating

$$\text{pMSE} = \text{bias}^2 + \text{Var}(\hat{Y}|X)$$

(prediction mean squared error) for the different covariate counts. We naturally want to minimize this value. The pMSE can't be computed directly as it uses the true model and therefore we use

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i^{new} - \hat{y}(m)_i)^2,$$

where y_i^{new} is the true value from the test set and $\hat{y}(m)_i$ is the predicted value for model m, as a replacement.

It is important to note that the "hack" we did to allow the categorical covariate while using *regsubsets* has the consequence that the categorical covariate is included in all models, meaning that the lowest number of allowed covariates is 2. The results of each of the models can be seen in figure 4 where we see that the model with the lowest $\sqrt{\text{pMSE}}$ is the model with 7 covariates. However, the model with 3 covariates, corresponding to our previously chosen model (3), could be seen as the most parsimonious model and thus be the best choice still as it differs only by a small amount compared to the model with 7 covariates. The covariates selected in each of the models can be seen in table 2. Note that not all continuous covariates were allowed.

This method for testing the predictive ability of a model is dependent on which data that was selected for training and testing and is thus not fully robust. In order to weaken the dependency, we are interested in seeing the results when different data is in the test set. Thus we are interested in using methods such as K-fold cross validation or LOOCV (leave one out cross validation). We explore this further in the results.

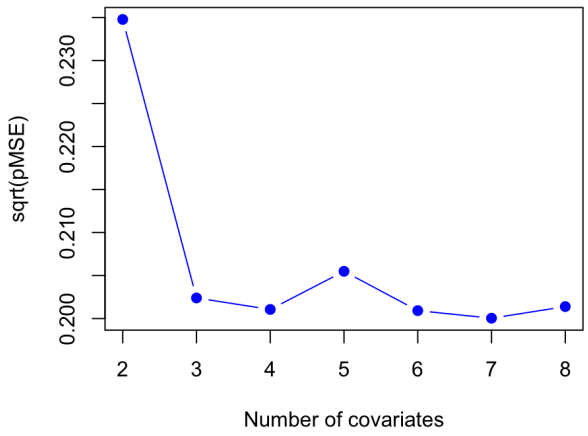


Figure 4: $\sqrt{\text{pMSE}}$ on test data for models with different covariate amounts. Only the models with lowest $\sqrt{\text{pMSE}}$ were kept for each covariate amount.

Model	*carbody*	*curbweight*	*horsepower*	*carlength*	*boreratio*	*carwidth*	*citympg*	*enginesize*
2	X	X	-	-		-	-	-
3	X	X	X	-	-	-	-	-
4	X	X	X	X	-	-	-	-
5	X	X	X	X	X	-	-	-
6	X	X	X	X	X	X	-	-
7	X	X	X	X	X	X	X	-
8	X	X	X	X	X	X	X	X

Table 2: Covariates selected (marked with X) in each model from exhaustive regsubsets, only a subset of the continuous covariates were selected.

Something which may be of interest is the interaction between covariates. Intuitively a interaction might be relevant between *curbweight* and *horsepower* as horsepower might increase the price at a higher rate for heavier cars. Likewise, an interaction between the covariates *horsepower* and *carbody* could be interesting as horsepower could be valued differently for certain car bodies. We find that while the interactions, specifically between *horsepower* and *carbody*, could occur at a statistically significant level, incorporating interactions into our model does not improve the predictive performance when using LOOCV (see Appendix 4). Thus we remain with the simpler model (3) while noting that interaction effects could still be relevant depending on what is desired of the model.

We now are ready to establish and interpret the results for the final model (3).

2.2 Results

We start by interpreting the model parameters. As the intercept normally would be interpreted as the log price of a hatchback car with *curbweight* $= 0$ and *horsepower* $= 0$, which is not particularly realistic, we are interested in first centering the covariates. In table 3 we see the estimated coefficients of the model now. The centered intercept $\beta_0 = 9.291$ can be interpreted as the expected log price for a hatchback car with average weight and horsepower (relative to our data). Thus the interpretation of the intercept is $\exp(9.291) \approx 10840$ which would be the price (in USD) of an hatchback with average weight and horsepower.

The coefficient for *curbweight* $\beta_1 = 0.0006$ can be interpreted as the expected percentage increase of the price for each additional pound of weight added to the car, above the mean, while the other coefficients are constant. This can be calculated by

$$100 \cdot (e^{0.0006} - 1) \approx 0.06\% \text{ increase in price per additional pound.}$$

Similarly, the coefficient for *horsepower* $\beta_2 = 0.0043$ can be interpreted in the same way. By performing the same calculation we get

$$100 \cdot (e^{0.0043} - 1) \approx 0.43\% \text{ increase in price per additional horsepower.}$$

Meaning that for each additional horsepower the price increases by approximately 0.43%, where we again assume that the other covariates are held constant and that the horsepower is above the mean.

For the categorical covariate *carbody*, the coefficients are interpreted relative the the baseline level *hatchback* where the weight and horsepower of the car is average. The expected percentage price change compared to a hatchback for the other levels are then

- *convertible*: $\beta_{convertible} = 0.335 \implies 100 \cdot (e^{0.335} - 1) \approx 40\%$ increase in price
- *hardtop*: $\beta_{hardtop} = 0.335 \implies 100 \cdot (e^{0.187} - 1) \approx 20.5\%$ increase in price
- *sedan*: $\beta_{sedan} = 0.335 \implies 100 \cdot (e^{0.112} - 1) \approx 11.9\%$ increase in price
- *wagon*: $\beta_{wagon} = 0.335 \implies 100 \cdot (e^{-0.048} - 1) \approx 4.7\%$ decrease in price.

Covariate	Estimated parameter
intercept	9.291
curbweight	0.0006
horsepower	0.0043
convertible	0.335
hardtop	0.187
sedan	0.112
wagon	-0.048

Table 3: Estimated coefficients of model (3) with centered covariates.

Naturally, we are now interested in the confidence intervals of our coefficients for our model (3) with centered covariates. Using same methods of interpretation as previously we can read from table 4 that the 95% confidence intervals provide a range of values for each coefficient, giving us an idea of the uncertainty in our estimates.

For the intercept, the 95% confidence interval tells us that the expected price of a average hatchback car should with 95% confidence be within the interval [10384 USD, 11339 USD].

Next, for the continuous covariates, we find that the 95% confidence interval for *curbweight* represents the price increase per additional pound of weight (above the mean), which ranges from 0.052% to 0.068%. For *horsepower* we have a percentage price increase per additional horsepower (above the mean) between 0.325% and 0.533% with 95% confidence.

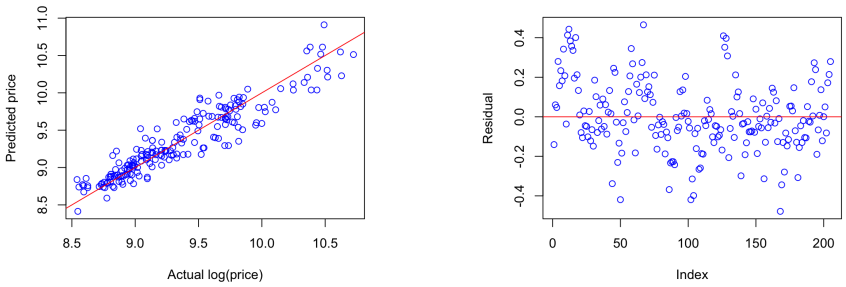
For the categorical covariate *carbody*, the 95% confidence intervals give us a measure on the range of price changes relative to a hatchback car with average weight and horsepower. We have

- $convertible : \beta_{convertible} \in [0.182, 0.487] \implies$ 20% to 63% increase in price
- $hardtop : \beta_{hardtop} \in [0.053, 0.322] \implies$ 5.4% to 38% increase in price
- $sedan : \beta_{sedan} \in [0.053, 0.171] \implies$ 5.4% to 19% increase in price
- $wagon : \beta_{wagon} \in [-0.139, 0.043] \implies$ 13% decrease to 4.4% increase in price. Note that this interval includes 0 and as such the effect is not statistically significant at the 5% level.

Covariate	2.5%	97.5%
Intercept	9.246	9.336
Curb weight	0.00052	0.00068
Horsepower	0.00324	0.00532
Convertible	0.182	0.487
Hardtop	0.053	0.322
Sedan	0.053	0.171
Wagon	-0.139	0.043

Table 4: 95% confidence intervals for regression coefficients for model (3) with centered covariates.

We look at some diagnostics for the model. Fitting the model on the full data set and plot the observed price vs the predicted price we get figure 5a. The fitted values appear to have a strong linear relation to the observed true values. This is also demonstrated in the residual plot in figure 5b, where the residuals appear symmetric around 0 and do not display any particular patterns. In the histogram of the residuals in figure 6 we see that the residuals follow a near symmetric Gaussian distribution with mean 0.



(a) Fitted vs actual values.

(b) Residuals from our model.

Figure 5: Model diagnostics.

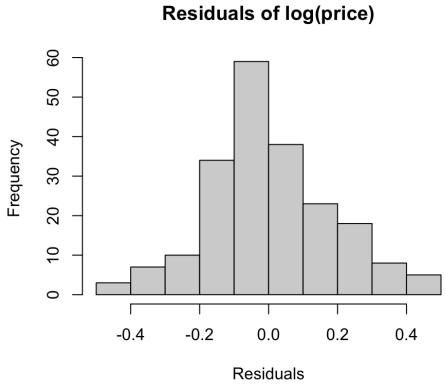


Figure 6: Histogram of residual counts.

The predictive ability of the model is assessed via LOOCV as the data set is quite small with only 205 observations. Thus we fit the model 205 times, each time leaving out a single observation for cross validation. The result of the model performance can be seen in table 5. The LOOCV rMSE is 0.185 on the log scale, meaning that the predicted prices are typically within $\pm 100 \cdot (e^{0.185} - 1) \approx \pm 20\%$ of the true price. The mean LOOCV error $y^{new} - \hat{y}$ is near zero, which indicates that the model is unbiased and that errors are not skewed. Although a few extreme observations have larger prediction errors, as can be seen in the minimum and maximum error being -0.548 and 0.476 respectively.

Metric	Value
Minimum error	-0.548
Maximum error	0.476
Mean error	-0.0001
Median error	-0.023
Standard deviation	0.186
LOOCV rMSE	0.185

Table 5: Summary of LOOCV prediction errors for the model (3).

We conclude this section by commenting that the R^2 for the model on the full data set is $R^2 = 0.8766$.

2.3 Discussion

The results indicate that the covariates *curbweight* and *horsepower* are strong predictors of car price. When looking at a arbitrary car body type and letting the horsepower be constant we have observed that heavier cars are strongly associated with higher prices, with an estimated price increase of almost 0.05% to 0.07% per additional pound above the mean. Although this effect is small for an individual pound increase it quickly has a substantial cumulative effect on the price. Similarly have we seen that horsepower has a large impact on price, one additional horsepower in a car is associated with an approximately 0.3% to 0.5% increase in price. This highlights that engine performance is valued highly in the car market.

We have also seen that *carbody* has a pronounced effect on price. Convertibles, hardtops and sedans are all priced significantly higher than hatchbacks where for example convertibles are priced 20 to 63% higher compared to a hatchback with the same weight and horsepower. On the other hand, the effect of a car being a wagon compared to a hatchback is smaller and more uncertain, as is reflected in the 95% confidence interval containing 0. Thus suggesting that, while taking car weight and horsepower into account, the price of a wagon compared to a hatchback is not significantly different.

The narrow confidence intervals for *curbweight* and *horsepower* indicate that the effect for these covariates are estimated with high precision. Meanwhile the larger confidence intervals for the different levels of *carbody* indicate a higher variance among this covariate. This could be caused by certain car body types having fewer observations, leading to higher uncertainty.

The diagnostic plots show a strong linear relation between fitted values and observed values, this indicates that our log transformation of the price response is justified. We also see that the basic assumptions are reasonably fulfilled as the residuals appear approximately symmetrical around 0 and show no clear patterns.

The high coefficient of determination $R^2 = 0.8766$ indicates that our model explains a large portion of the total variance of the car prices in the data. We also found strong results for the predictive ability of the model using LOOCV

where the mean error was near 0 for the cross validations. There were however some extreme observations which led to larger errors. This could be attributed to the fact that we did not previously remove outliers which our model could not explain as our model's considered covariates do not capture the variance in these outlier data points.

2.4 Summary

In our analysis we used many diagnostic as well as algorithmic methods as means to find meaningful covariates for the multivariate linear model which we used to describe the observed price in the data. We used a parsimonious model with covariates *curbweight*, *horsepower* and *carbody* with a log transformed response covariate, which explains 87.7% of the variance in the price of the cars in the data ($R^2 = 0.8766$) and performs well in predicting outside of the data with LOOCV rMSE≈ 0.185, corresponding to predictions typically being within $\pm 20\%$ of the true price. Still, the model does not capture the price variation caused by for example brands and rare configurations such as rear-engined cars and thus has limitations in generalizing to premium or unusual vehicles.

The covariates *curbweight* and *horsepower* have large cumulative effects, each additional pound or horsepower increase while the other covariates are held constant lead to an estimated price increase of 0.06% and 0.43% respectively. The categorical covariate *carbody* was also shown to be significant as for example convertibles are estimated as approximately 40% more expansive than hatchbacks.

3 Data on GLMs: operational errors by air-traffic controllers

The controllers.dat file contains the 12 observations where each observation has a unique age and experience combination(Each age group 1 & 2 at each level of experience, 1-6). The sample size varies a lot for each observation which indicates that the model should be weighted with size. The response co variate is *error rate* which is the number of errors a member of a certain observation is likely to make,

$$ErrorRate = \frac{Error}{Size}. \tag{4}$$

The covariates are converted into categorical covariates where experience level 5 and age level 1 are set as the baseline. The goal is to determine the rate of errors based on some sample size.

3.1 Methods and model

By using a binomial GLM with weights equivalent to the size of each group, it is equivalent to modeling the errors even though the error rate is the response.

The predicted errors in each group based on its sample size is given as

$$error_i \sim Binomial(size_i, p_i) \tag{5}$$

where p_i is the probability that any given person or "trial" in group $i = 1, \ldots, 12$ will make an error. This is interpreted as the error rate which we want the model to predict. Since size will be used as weights in the model, it is safe to assume that the groups with larger sizes will better fit the model. The function used to link the model to the binomial distribution is a logit function,

$$log\left(\frac{p_i}{1-p_i}\right) \tag{6}$$

which will ensure that the responses, predicted error rates, are between 0 and 1. The GLM can now be constructed from the categorical covariates experience and age. The model constructed is

$$log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1 I(exp = 1) + \beta_2 I(exp = 2) + \beta_3 I(exp = 3) + \beta_4 I(exp = 4) + \beta_5 I(exp = 6) + \beta_6 I(age = 2). \tag{7}$$

The model has experience level 5 and age level 1 as the baseline, they are incorporated into the intercept.

3.2 Results and discussion

The resulting model gives the following covariate coefficients:

Coefficient	Value
β_0	-2.73697
β_1	-1.58926
β_2	0.06045
β_3	0.16202
β_4	0.11291
β_5	-0.09936
β_6	-0.17393

Table 6: Coefficient values of the GLM 7.

The coefficients in table 6 shows the predicted error rates per group. β_0 (Baseline of experience level 5 and age level 1) gives a predicted error of

$$p_i = \frac{e^{-2.73697}}{1 + e^{-2.73697}} \approx 0.061 = 6.1\% \tag{8}$$

The other parameters are representations for the other groups that are not the baseline. The following table 7 shows the predicted error rate for each group:

Experience	Age	Change from baseline error rate
1	1	-4.8%
1	2	-5.0%
2	1	+0.3%
2	2	-0.63%
3	1	+0.91%
3	2	-0.1%
4	1	+0.66%
4	2	-0.30%
5	1	0 (Baseline)
5	2	-0.94%
6	1	-0.56%
6	2	-1.4%

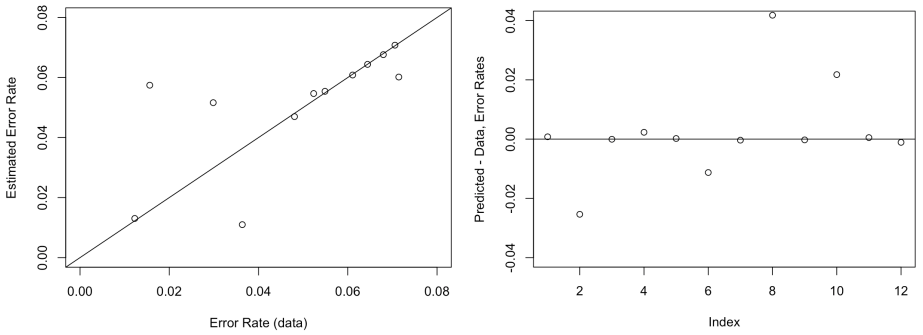
Table 7: Change of error rate from baseline for each group

Experience	Age	Size	Difference of error rate
1	1	3587	0.078%
1	2	110	2.53%
2	1	7574	0.0058%
2	2	191	-0.23%
3	1	15758	-0.02%
3	2	280	1.1%
4	1	14816	0.036%
4	2	128	-4.2%
5	1	5615	0.026%
5	2	67	-2.2%
6	1	2587	-0.05%
6	2	1186	-0.11%

Table 8: Difference of error rate between predicted and data

Analyzing the error rates for the different groups some patterns emerge. Age level 2 has lower error rates from baseline while Age level 1 has higher error rates. The error rates for the lowest experience level are much lower than the baseline. This could be due to less experienced ATCs being stationed at airports which are not as busy and are therefore subject to a less stressful environment. Experience level 6 has lower error rates compare to the baseline, this could be partly explained by the broad interval of this group. The group includes all individuals with ≥ 25 years of experience, possibly leading to some cases with extreme experience time and thus reducing the error rate.

The predicted error rates are considerably better for groups with larger sizes, contains more data. These data points are most likely more reliable as they are a closer representation to the population. The model is also constructed with size as weights so it will naturally be better fit for data groups with bigger sizes.



(a) Predicted Error rates vs Data Error rates

(b) Residuals of Predicted vs Data Error rates

Figure 7: Predicted error rates and residuals from GLM constructed

4 Final remarks and take home messages

The two sections of the project were unrelated where the first involved a larger systematic method for finding a best fit linear regression model. Transformation of data, hypothesis testing and backward search were all involved in finding the model. The efforts resulted in a model which involved the fewest covariates while maintaining a good fit for the data which showed well distributed residuals without biasing and a high R^2 value.

The second part of the project, was more condensed with simply creating a general linear model (GLM) of the given data. Looking at the data, it made most sense to use a Binomial GLM where the odds of a single individual committing an error. The Binomial GLM which was created, behaved as expected where data points of larger size were more influential in fitting the model. Higher age group (age=2) had lower error rates and that lower experience on the whole had higher error rates. This is not very consistent as the group sizes varied a lot so some predicted error rates were far from the given error rates the data showed.

The take home from us is that regression modeling can have very different applications and different approaches depending on the data given to fit a model. A lot of work in R is required to best determine and understand the covariates and their affect on the predicted outcome.

Appendix A: Car data

Redundancy in *citympg* and *highwaympg*

In figure 8 we see the R output which shows the redundancy of including both *citympg* and *highwaympg* in the model as they are highly correlated.

```
> cor(car_price_data$citympg, car_price_data$highwaympg)
[1] 0.971337
```

Figure 8: R output showing high correlation between covariates *citympg* and *highwaympg*

Deciding on continuous covariates for smaller model

Considering the AIC and C(p) values, it appears that having the three covariates *curbweight*, *horsepower* and *carwidth* will give the best model for prediction without being over-complex as can be seen in the OLSRR analysis in figure 12.

The variance inflation factor analysis of this model in figure 10 demonstrates multicollinearity among the covariates as the highest factor is 5.45 for *curbweight*. This is not unexpected as the weight of a car should be highly correlated to its size (e.g. width). We try removing *carwidth* as this covariate does not seem to describe the price of a car as well as *curbweight* does. This reduces the VIF to a much more acceptable 2.29.

To further improve the model we consider the addition of fuel consumption covariates. Figure 9 shows that including *citympg* (and *highwaympg* but we have shown both to be redundant) in model 5 yields the lowest MSEP and a slightly better R^2 and thus may explain a portion of the variance which *curbweight* and *horsepower* fails to capture. We see that adding *citympg* to the model does not cause problematic multicollinearity as the highest value can be seen in figure 10 to be 3.31 for *citympg*.

Best Subsets Regression

Model Index	Predictors
1	curbweight
2	curbweight horsepower
3	carwidth curbweight horsepower
4	carlength carwidth curbweight horsepower
5	carwidth curbweight horsepower citympg highwaympg
6	carwidth curbweight enginesize horsepower citympg highwaympg
7	carlength carwidth curbweight enginesize horsepower citympg highwaympg
8	carlength carwidth curbweight enginesize boreratio horsepower citympg highwaympg
9	wheelbase carlength carwidth curbweight enginesize boreratio horsepower citympg highwaympg

Subsets Regression Summary

Model	R-Square	Adj. R-Square	Pred R-Square	C(p)	AIC	SBIC	SBC	MSEP	FPE	HSP	APC
1	0.7943	0.7933	0.7903	84.6922	-18.4965	-601.5506	-8.5274	10.7558	0.0530	3e-04	0.2098
2	0.8506	0.8491	0.8433	8.4929	-82.0592	-663.8963	-68.7671	7.8506	0.0389	2e-04	0.1538
3	0.8551	0.8530	0.8445	4.2191	-86.3532	-667.9684	-69.7381	7.6513	0.0380	2e-04	0.1506
4	0.8561	0.8532	0.8442	4.8465	-85.7564	-667.2647	-65.8184	7.6373	0.0382	2e-04	0.1511
5	0.8575	0.8539	0.8424	4.9336	-85.7281	-667.0670	-62.4670	7.6024	0.0382	2e-04	0.1511
6	0.8591	0.8548	0.8424	4.7158	-86.0380	-667.1432	-59.4539	7.5553	0.0381	2e-04	0.1509
7	0.8595	0.8545	0.8416	6.0889	-84.6957	-665.6528	-54.7886	7.5696	0.0384	2e-04	0.1519
8	0.8596	0.8539	0.8398	8.0075	-82.7813	-663.6290	-49.5512	7.6052	0.0387	2e-04	0.1533
9	0.8596	0.8531	0.8376	10.0000	-80.7892	-661.5336	-44.2360	7.6441	0.0391	2e-04	0.1548

Figure 9: OLSRR analysis with continuous variables

```
> vif(lm(log(price)~curbweight+horsepower+carwidth,data=car_price_data))
curbweight horsepower   carwidth
  5.446270   2.293721   4.031980
> vif(lm(log(price)~curbweight+horsepower,data=car_price_data))
curbweight horsepower
  2.291525   2.291525
> vif(lm(log(price)~curbweight+horsepower+citympg,data=car_price_data))
curbweight horsepower    citympg
  2.713072   3.233861   3.310218
```

Figure 10: The variance inflation factor of the best 3 covariate model returned from OLSRR, the same model with *carwidth* removed and a model with *curbweight, horsepower* and *citympg*.

Deciding on categorical covariate

We let OLSRR perform best subset selection allowing a selection of both continuous and categorical covariates to be picked. We see in figure 12 when both *curbweight* and *horsepower* is already in the model, the next added covariate is the categorical covaraite *carbody*.

The inclusion of *carbody* leads to noticeable improvement in model fit as R^2 increases from 0.851 to 0.877 and as such can be regarded as the most important categorical covariate and as such is included in our model.

Best Subsets Regression

Model Index	Predictors
1	curbweight
2	curbweight horsepower
3	curbweight horsepower carbody
4	curbweight horsepower cylindernumber carbody
5	curbweight horsepower cylindernumber carbody drivewheel
6	carwidth curbweight horsepower cylindernumber carbody drivewheel

Figure 11: Olsrr best subsets with continuous and catagorical covariates

Subsets Regression Summary

Model	R-Square	Adj. R-Square	Pred R-Square	C(p)	AIC	SBIC	SBC	MSEP	FPE	HSP	APC
1	0.7943	0.7933	0.7903	191.2212	-18.4965	-602.6263	-8.5274	10.7558	0.0530	3e-04	0.2098
2	0.8506	0.8491	0.8433	85.8628	-82.0592	-665.6633	-68.7671	7.8506	0.0389	2e-04	0.1538
3	0.8766	0.8729	0.8636	44.3033	-113.2413	-702.0696	-86.6572	6.5172	0.0329	2e-04	0.1283
4	0.8932	0.8866	-Inf	24.5640	-130.9448	-728.6392	-84.4227	5.6664	0.0295	1e-04	0.1121
5	0.8971	0.8895	-Inf	21.2121	-134.4856	-733.7169	-81.3174	5.4894	0.0289	1e-04	0.1091
6	0.9009	0.8930	-Inf	16.0000	-140.1626	-738.7566	-83.6715	5.3144	0.0281	1e-04	0.1061

Figure 12: Olsrr best subsets analysis with continuous and catagorical covariates

Outliers analysis

In order to analyze if outliers in our data set is can be disregarded or if they are representative to the vehicle market, we begin by looking at figure 13 and note some of the outliers. Some being observations $(12, 67, 168)$.

- In figure 14 for observation 12 we see that this car is more expansive than other cars with similar *horsepower* and *curbweight*. Therefore this is marked as a outlier with respect to our model as it only considers these covariates and the *carbody* of the vehicle. But the increase of price in the observation could be explained because of the premium brand of the vehicle (BMW) and thus it is not a data error and we should thus keep it in the data.

- Observation 67 again has a very high *price* compared to its *horsepower* which could be explained by its high *compressionratio* which our model fails to capture. Again this should be seen as a shortcoming of our model and not be removed.

- In observation 168 we see that this vehicle instead has a cheap *price* relative to *horsepower* which could be a effect of the brand being a cheaper brand (Toyota) which our model does not consider. Thus we keep the observation in our data set.

We also note of the outliers $(127, 128, 129)$ which we can read are the only 3 observations where the *enginelocation* is 'rear'. These could be seen as extreme cases and thus be removed as the infliction of this change will be extreme compared to otherwise similar vehicles. But as these observations are not errors, just

rather unusual configurations in vehicles we decide to keep them as removing them would be making the data more biased towards more "average" cars.

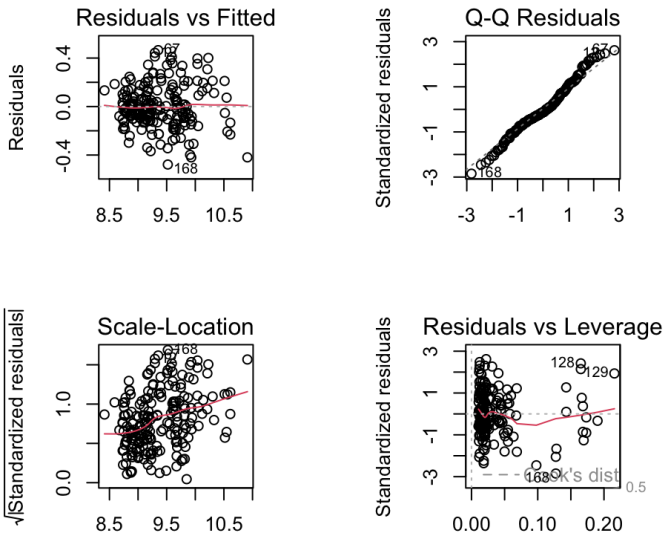


Figure 13: Plot of regression models using model (3)

```
> car_price_data[12,]
   car_ID symboling  CarName fueltype aspiration doornumber carbody drivewheel enginelocation wheelbase carlength
12     12         0 bmw 320i      gas        std       four   sedan        rwd          front     101.2     176.8
   carwidth carheight curbweight enginetype cylindernumber enginesize fuelsystem boreratio stroke compressionratio
12     64.8      54.3       2395        ohc           four        108       mpfi       3.5    2.8              8.8
   horsepower peakrpm citympg highwaympg price
12        101    5800      23         29 16925
> car_price_data[67,]
   car_ID symboling       CarName fueltype aspiration doornumber carbody drivewheel enginelocation wheelbase carlength
67     67         0 mazda rx-7 gs   diesel        std       four   sedan        rwd          front     104.9       175
   carwidth carheight curbweight enginetype cylindernumber enginesize fuelsystem boreratio stroke compressionratio
67     66.1      54.4       2700        ohc           four        134        idi      3.43   3.64             22
   horsepower peakrpm citympg highwaympg price
67         72    4200      31         39 18344
> car_price_data[168,]
    car_ID symboling               CarName fueltype aspiration doornumber carbody drivewheel enginelocation wheelbase
168    168         2 toyota corona liftback      gas        std        two hardtop        rwd          front      98.4
    carlength carwidth carheight curbweight enginetype cylindernumber enginesize fuelsystem boreratio stroke
168     176.2      65.6        52       2540        ohc           four        146       mpfi      3.62    3.5
    compressionratio horsepower peakrpm citympg highwaympg price
168              9.3        116    4800      24         30  8449
```

Figure 14: Selected observations from the car price data set

Interaction analysis

We perform a partial F-test between our regular model (3) and one incorporating the interaction between *curbweight* and *horsepower* and we see in figure 15 that we cannot reject the interaction as it is statistically significant with $p = 0.042$. In a similar way, we compare our simple model (3) to the model with interaction between *horsepower* and *carbody* and see in figure 16 an even higher

statistical significance with $p = 0.002$ and again cannot reject the model with the interaction. This is consistent with the differing real world market evaluations based not only of performance of a car but also of its appearance.

We look further by comparing the predictive ability of the models with interaction compared to the one without. We see in table 9 that although interaction between covariates were significant in the partial F-tests, they do not marginally improve the predictive performance when using LOOCV. LOOCV rMSE values were for example almost identical across the models and similar results were found for the median and mean error.

```
Analysis of Variance Table

Model 1: log(price) ~ curbweight + horsepower + carbody
Model 2: log(price) ~ curbweight * horsepower + carbody
  Res.Df    RSS Df Sum of Sq      F  Pr(>F)
1    198 6.3897
2    197 6.2566  1   0.13312 4.1914 0.04196 *
```

Figure 15: Partial F-test between model with no interaction and one with interaction between covariates *curbweight* and *horsepower*.

```
Analysis of Variance Table

Model 1: log(price) ~ curbweight + horsepower + carbody
Model 2: log(price) ~ curbweight + horsepower * carbody
  Res.Df    RSS Df Sum of Sq      F   Pr(>F)
1    198 6.3897
2    194 5.8561  4   0.53356 4.4189 0.001931 **
```

Figure 16: Partial F-test between model with no interaction and one with interaction between covariates *horsepower* and *carbody*.

Metric	No interaction	*curbweight*horsepower*	*horsepower*carbody*
Minimum error	-0.548	-0.566	-0.487
Maximum error	0.476	0.467	0.645
Mean error	-0.0001	-0.0002	0.002
Median error	-0.023	-0.023	-0.030
Standard deviation	0.186	0.185	0.186
LOOCV rMSE	0.185	0.185	0.185

Table 9: Summary of LOOCV prediction errors for three models. One with no interactions, one with interaction between covariates *curbweight* and *horsepower* and one between *horsepower* and *carbody*.

Appendix B: GLM data

ATC Errors

The summary of the shows that both categorical variables are significant and valid to be used. The residual deviance for the model is 11.4.

```
Call:
glm(formula = error_rate ~ experience + factor(age), family = binomial(link = "logit"),
    data = df, weights = size)

Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept)  -2.73697    0.05556 -49.258   <2e-16 ***
experience1  -1.58926    0.15555 -10.217   <2e-16 ***
experience2   0.06045    0.07234   0.836   0.4034
experience3   0.16202    0.06354   2.550   0.0108 *
experience4   0.11291    0.06441   1.753   0.0796 .
experience6  -0.09936    0.09685  -1.026   0.3050
factor(age)2 -0.17393    0.11572  -1.503   0.1329
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 269.077  on 11  degrees of freedom
Residual deviance:  11.434  on  5  degrees of freedom
AIC: 93.938

Number of Fisher Scoring iterations: 4
```

Figure 17: Summary of model

The residual deviance is used to test the binomial distribution, which is assumed for the model. If the variance of the data is larger than what the model assumes it is, the model is not that good at capturing the variance and the overdispersion is high. For this project, the data is largely heterogeneous sizes, few data points and aggregate errors which will lead to the model not being good at capturing the full variance of the data. The overdispersion is calculated to 2.28 according to the figure below. A overdispersion value of 1 is

```
> deviance(fit_no_int_2)/df.residual(fit_no_int_2)
[1] 2.286817
```

Figure 18: Overdispersion of the model

desired, above this and the model is not good at capturing the variance of the data. However, considering the data available it is difficult to make any better models. The residuals of the models shows that the lack of size in some of the data points leads to high variance which is difficult to capture.